

Data Analytics

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Course : Data Analytics

Subject : SQL

Batch Time : 8 to 9

1. 9) Explain normalization in the context of databases.

- **Normalization** is a way of **organizing data in a database** so that:

- data is **not repeated (no duplication)**
- data is **stored neatly in separate tables**
- database becomes **easy to manage**

- Ex

- **Student Table**

- student_id	student_name
- 1	Rahul

- **Course Table**

course_id	course_name
- 101	Math
- 102	Science

- **Student_Course Table**

- student_id	course_id
- 1	101
- 1	102

2. 10) What does a join in MySQL mean?

- A **JOIN** is used to **combine data from two or more tables**.
- It connects tables using a **common column** (like ID).
- "JOIN means taking related data from different tables and showing it together."

3. 11) What do you understand about DDL, DCL, and DML in MySQL?

1. DDL (Data Definition Language)

DDL is used to **create and manage tables (structure of database)**

Related to **table structure**

Used to **create, change, or delete tables**

Commands:

CREATE → create table

ALTER → change table

DROP → delete table

2. DML (Data Manipulation Language)

DML is used to **work with data inside tables**

Related to **data**

Used to **insert, update, and delete data**

Commands:

INSERT → add data

UPDATE → change data

DELETE → remove data

Example:

```
INSERT INTO table_name (col1, col2) VALUES (value1, value2);
```

3. DCL (Data Control Language)

DCL is used to control access and permissions

Commands:

GRANT → give permission

REVOKE → remove permission

4 . 12) What is the role of the MySQL JOIN clause in a query, and what are some common types of joins?

A **JOIN** is used to **combine data from 2 or more tables**.

It works using a **common field (column)** between tables.

JOIN connects tables and shows related data together.

1. INNER JOIN

Shows only **matching data in both tables**

2. LEFT JOIN

Shows **all data from left table** + matching data from right table

3. RIGHT JOIN

Shows **all data from right table** + matching data from left table

4. FULL JOIN (not directly in MySQL but concept)

Shows **all data from both tables**

JOIN is used to combine data from two or more tables using a common field.

Common types are INNER JOIN, LEFT JOIN, and RIGHT JOIN.